Machine Learning Exercise 2A

Problem 1 *In this problem, you will implement a logistic regression model to predict whether a student gets admitted into a university. Suppose that you are the administrator of a university department and you want to determine each applicant's chance of admission based on their results on two exams. You have historical data from previous applicants that you can use as a training set for logistic regression. For each training example, you have the applicant's scores on two exams and the admissions decision. Your task is to build a classification model that estimates an applicant's probability of admission based on the scores from those two exams.*
The file `ex2data1.txt` *contains the dataset.*
Consider the hypothesis function corresponding to the logistic model:

$$h_\theta(x) = g(\theta^T x),$$

where g is the sigmoid function, which is defined as:

$$g(z) = \frac{1}{1+e^{-z}}.$$

i) Plot the data so that it displays a figure where the axes are the two exam scores, and the positive and negative examples are shown with different markers.

ii) Implement the sigmoid function.

iii) Implement a function to calculate the cost:

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log(h_\theta(x^{(i)})) + (1-y^{(i)})\log(1-h_\theta(x^{(i)}))],$$

and the gradient:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})x_j^{(i)}.$$

iv) Optimise the cost function with the `minimize` *function from* `scipy.optimize`*, using the cost and gradient functions as inputs. Print the optimal values of θ.*

v) Check if you get the same results using `scikit-learn` *or GLM.*

vi) Plot the decision boundary.

vii) Use the model to predict whether a student with an Exam 1 score of 45 and an Exam 2 score of 85, will be admitted.

viii) Report the training accuracy of your classifier by computing the percentage of examples it got correct.